

UNIT – II

LONGITUDINAL VIBRATIONS

FORMULAE:

Time period, $t_p = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{s}}$

natural frequency, $f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{s}{m}} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$

Taking the value of g as 9.81 m/s^2 and δ in metres,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{9.81}{\delta}} = \frac{0.4985}{\sqrt{\delta}} \text{ Hz}$$

when the mass of the constraint m_C and the mass of the disc m at the end is given, then natural frequency of vibration,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{s}{m + \frac{m_C}{3}}}$$

Frequency of Free Damped Vibrations (Viscous Damping):

$$\frac{s}{m} = (\omega_n)^2$$

$$\omega_d = \sqrt{\frac{s}{m} - \left(\frac{c}{2m}\right)^2} = \sqrt{(\omega_n)^2 - a^2} \quad ; \text{ and } a = \frac{c}{2m}$$

We know that the periodic time of vibration,

$$t_p = \frac{2\pi}{\omega_d} = \frac{2\pi}{\sqrt{\frac{s}{m} - \left(\frac{c}{2m}\right)^2}} = \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

and frequency of damped vibration,

$$f_d = \frac{1}{t_p} = \frac{\omega_d}{2\pi} = \frac{1}{2\pi}\sqrt{(\omega_n)^2 - a^2} = \frac{1}{2\pi}\sqrt{\frac{s}{m} - \left(\frac{c}{2m}\right)^2}$$

When there is no damper,

$$f_n = \frac{1}{2\pi} \sqrt{\frac{s}{m}}$$

Damping factor

$$= \frac{c}{c_c} = \frac{c}{2m\omega_n}$$

$$\dots (\because c_c = 2\pi\omega_n)$$

amplitude reduction factor,

$$\frac{x_1}{x_2} = \frac{Ae^{-at}}{Ae^{-a(t+t_p)}} = e^{at_p} = \text{constant}$$

Logarithmic decrement:

$$\delta = \log \left(\frac{x_1}{x_2} \right) = \log e^{at_p}$$

$$\delta = \log_e \left(\frac{x_1}{x_2} \right) = at_p = a \times \frac{2\pi}{\omega_d} = \frac{a \times 2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

$$\dots \left[\because \omega_d = \sqrt{(\omega_n)^2 - a^2} \right]$$

$$= \frac{2\pi \times c}{\sqrt{(c_c)^2 - c^2}}$$

In general, amplitude reduction factor,

$$\frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \dots = \frac{x_n}{x_{n+1}} = e^{at_p} = \text{constant}$$

$\therefore$ Logarithmic decrement,

$$\delta = \log_e \left(\frac{x_n}{x_{n+1}} \right) = at_p = \frac{2\pi \times c}{\sqrt{(c_c)^2 - c^2}}$$

PROBLEMS ON LONGITUDINAL VIBRATIONS:

1. A vibrating system consists of a mass of 200 kg, a spring of stiffness 80 N/mm and a damper with damping coefficient of 800 N/m/s. Determine the frequency of vibration of the system.

Solution. Given : $m = 200 \text{ kg}$; $s = 80 \text{ N/mm} = 80 \times 10^3 \text{ N/m}$; $c = 800 \text{ N/m/s}$

We know that circular frequency of undamped vibrations,

$$\omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{80 \times 10^3}{200}} = 20 \text{ rad/s}$$

and circular frequency of damped vibrations,

$$\omega_d = \sqrt{(\omega_n)^2 - a^2} = \sqrt{(\omega_n)^2 - (c/2m)^2} \quad \dots (\because a = c/2m)$$

$$= \sqrt{(20)^2 - (800/2 \times 200)^2} = 19.9 \text{ rad/s}$$

$\therefore$ Frequency of vibration of the system,

$$f_d = \omega_d / 2\pi = 19.9 / 2\pi = 3.17 \text{ Hz} \quad \text{Ans.}$$

2. The following data are given for a vibratory system with viscous damping: Mass = 2.5 kg ; spring constant = 3 N/mm and the amplitude decreases to 0.25 of the initial value after five consecutive cycles. Determine the damping coefficient of the damper in the system.

Solution. Given : $m = 2.5 \text{ kg}$; $s = 3 \text{ N/mm} = 3000 \text{ N/m}$; $x_6 = 0.25 x_1$

We know that natural circular frequency of vibration,

$$\omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{3000}{2.5}} = 34.64 \text{ rad/s}$$

Let c = Damping coefficient of the damper in N/m/s,

x_1 = Initial amplitude, and

x_6 = Final amplitude after five consecutive cycles = $0.25 x_1$...(Given)

We know that

$$\frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \frac{x_4}{x_5} = \frac{x_5}{x_6}$$

or
$$\frac{x_1}{x_6} = \frac{x_1}{x_2} \times \frac{x_2}{x_3} \times \frac{x_3}{x_4} \times \frac{x_4}{x_5} \times \frac{x_5}{x_6} = \left(\frac{x_1}{x_2} \right)^5$$

$$\therefore \frac{x_1}{x_2} = \left(\frac{x_1}{x_6} \right)^{1/5} = \left(\frac{x_1}{0.25 x_1} \right)^{1/5} = (4)^{1/5} = 1.32$$

We know that

$$\log_e \left(\frac{x_1}{x_2} \right) = a \times \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

$$\log_e(1.32) = a \times \frac{2\pi}{\sqrt{(34.64)^2 - a^2}} \quad \text{or} \quad 0.2776 = \frac{a \times 2\pi}{\sqrt{1200 - a^2}}$$

Squaring both sides,

$$0.077 = \frac{39.5 a^2}{1200 - a^2} \quad \text{or} \quad 92.4 - 0.077 a^2 = 39.5 a^2$$

$$\therefore a^2 = 2.335 \quad \text{or} \quad a = 1.53$$

We know that $a = c / 2m$ or $c = a \times 2m = 1.53 \times 2 \times 2.5 = 7.65 \text{ N/m/s}$ **Ans.**

3. An instrument vibrates with a frequency of 1 Hz when there is no damping. When the damping is provided, the frequency of damped vibrations was observed to be 0.9 Hz. Find 1. the damping factor, and 2. logarithmic decrement.

Solution. Given : $f_n = 1 \text{ Hz}$; $f_d = 0.9 \text{ Hz}$

1. Damping factor

Let m = Mass of the instrument in kg,

c = Damping coefficient

or damping force per unit

velocity in N/m/s, and

c_c = Critical damping coefficient in N/m/s.

We know that natural circular frequency of undamped vibrations

$$\omega_n = 2\pi \times f_n = 2\pi \times 1 = 6.284 \text{ rad/s}$$

and circular frequency of damped vibrations,

$$\omega_d = 2\pi \times f_d = 2\pi \times 0.9 = 5.66 \text{ rad/s}$$

We also know that circular frequency of damped vibrations (ω_d),

$$5.66 = \sqrt{(\omega_n)^2 - a^2} = \sqrt{(6.284)^2 - a^2}$$

Squaring both sides,

$$(5.66)^2 = (6.284)^2 - a^2 \text{ or } 32 = 39.5 - a^2$$

$$\therefore a^2 = 7.5 \quad \text{or} \quad a = 2.74$$

$$\text{We know that,} \quad a = c/2m \quad \text{or} \quad c = a \times 2m = 2.74 \times 2m = 5.48 \text{ m N/m/s}$$

$$\text{and} \quad c_c = 2m\omega_n = 2m \times 6.284 = 12.568 \text{ m N/m/s}$$

$\therefore$ Damping factor,

$$c/c_c = 5.48m/12.568m = 0.436 \text{ Ans.}$$

2. Logarithmic decrement

We know that logarithmic decrement,

$$\delta = \frac{2\pi c}{\sqrt{(c_c)^2 - c^2}} = \frac{2\pi \times 5.48m}{\sqrt{(12.568m)^2 - (5.48m)^2}} = \frac{34.4}{11.3} = 3.04 \text{ Ans.}$$

4. The measurements on a mechanical vibrating system show that it has a mass of 8 kg and that the springs can be combined to give an equivalent spring of stiffness 5.4 N/mm. If the vibrating system have a dashpot attached which exerts a force of 40 N when the mass has a velocity of 1 m/s, find : 1. critical damping coefficient, 2. damping factor, 3. Logarithmic decrement, and 4. ratio of two consecutive amplitudes.

Solution. Given : $m = 8 \text{ kg}$; $s = 5.4 \text{ N/mm} = 5400 \text{ N/m}$

Since the force exerted by dashpot is 40 N, and the mass has a velocity of 1 m/s, therefore Damping coefficient (actual),

$$c = 40 \text{ N/m/s}$$

1. Critical damping coefficient

We know that critical damping coefficient,

$$c_c = 2m\omega_n = 2m \times \sqrt{\frac{s}{m}} = 2 \times 8 \sqrt{\frac{5400}{8}} = 416 \text{ N/m/s Ans.}$$

2. Damping factor

We know that damping factor

$$= \frac{c}{c_c} = \frac{40}{416} = 0.096 \text{ Ans.}$$

3. Logarithmic decrement

We know that logarithmic decrement,

$$\delta = \frac{2\pi c}{\sqrt{(c_c)^2 - c^2}} = \frac{2\pi \times 40}{\sqrt{(416)^2 - (40)^2}} = 0.6 \text{ Ans.}$$

4. Ratio of two consecutive amplitudes

Let x_n and x_{n+1} = Magnitude of two consecutive amplitudes,

We know that logarithmic decrement,

$$\delta = \log_e \left[\frac{x_n}{x_{n+1}} \right] \text{ or } \frac{x_n}{x_{n+1}} = e^\delta = (2.7)^{0.6} = 1.82 \text{ Ans.}$$

5. A mass suspended from a helical spring vibrates in a viscous fluid medium whose resistance varies directly with the speed. It is observed that the frequency of damped vibration is 90 per minute and that the amplitude decreases to 20 % of its initial value in one complete vibration. Find the frequency of the free undamped vibration of the system.

Solution. Given : $f_d = 90/\text{min} = 90/60 = 1.5 \text{ Hz}$

We know that time period,

$$t_p = 1/f_d = 1/1.5 = 0.67 \text{ s}$$

Let

x_1 = Initial amplitude, and

x_2 = Final amplitude after one complete vibration

$$= 20\% x_1 = 0.2 x_1$$

... (Given)

We know that

$$\log_e \left(\frac{x_1}{x_2} \right) = a t_p \quad \text{or} \quad \log_e \left(\frac{x_1}{0.2 x_1} \right) = a \times 0.67$$

$$\therefore \log_e 5 = 0.67 a \quad \text{or} \quad 1.61 = 0.67 a \quad \text{or} \quad a = 2.4 \quad \dots (\because \log_e 5 = 1.61)$$

We also know that frequency of free damped vibration,

$$f_d = \frac{1}{2\pi} \sqrt{(\omega_n)^2 - a^2}$$

or

$$(\omega_n)^2 = (2\pi \times f_d)^2 + a^2 \quad \dots (\text{By squaring and arranging})$$
$$= (2\pi \times 1.5)^2 + (2.4)^2 = 94.6$$

$$\therefore \omega_n = 9.726 \text{ rad/s}$$

We know that frequency of undamped vibration,

$$f_n = \frac{\omega_n}{2\pi} = \frac{9.726}{2\pi} = 1.55 \text{ Hz Ans.}$$

6. A coil of spring stiffness 4 N/mm supports vertically a mass of 20 kg at the free end. The motion is resisted by the oil dashpot. It is found that the amplitude at the beginning of the fourth cycle is 0.8 times the amplitude of the previous vibration. Determine the damping force per unit velocity. Also find the ratio of the frequency of damped and undamped vibrations.

Solution. Given : $s = 4 \text{ N/mm} = 4000 \text{ N/m}$; $m = 20 \text{ kg}$

Damping force per unit velocity

Let c = Damping force in newtons per unit velocity i.e. in N/m/s

x_n – Amplitude at the beginning of the third cycle,

x_{n+1} = Amplitude at the beginning of the fourth cycle = $0.8 x_n$
..... (Give

We know that natural circular frequency of motion,

$$\omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{4000}{20}} = 14.14 \text{ rad/s}$$

and

$$\log_e \left(\frac{x_n}{x_{n+1}} \right) = a \times \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

or

$$\log_e \left(\frac{x_n}{0.8 x_n} \right) = a \times \frac{2\pi}{\sqrt{(14.14)^2 - a^2}}$$

$$\log_e 1.25 = a \times \frac{2\pi}{\sqrt{200 - a^2}} \quad \text{or} \quad 0.223 = a \times \frac{2\pi}{\sqrt{200 - a^2}}$$

Squaring both sides

$$0.05 = \frac{a^2 \times 4\pi^2}{200 - a^2} = \frac{39.5a^2}{200 - a^2}$$

$$0.05 \times 200 - 0.05 a^2 = 39.5a^2 \quad \text{or} \quad 39.55 a^2 = 10$$

$$\therefore a^2 = 10 / 39.55 = 0.25 \quad \text{or} \quad a = 0.5$$

We know that

$$a = c / 2m$$

$$\therefore c = a \times 2m = 0.5 \times 2 \times 20 = 20 \text{ N/m/s} \quad \text{Ans.}$$

Ratio of the frequencies

$$\text{Let} \quad f_{n1} = \text{Frequency of damped vibrations} = \frac{\omega_d}{2\pi}$$

$$f_{n2} = \text{Frequency of undamped vibrations} = \frac{\omega_n}{2\pi}$$

$\therefore$

$$\frac{f_{n1}}{f_{n2}} = \frac{\omega_d}{2\pi} \times \frac{2\pi}{\omega_n} = \frac{\omega_d}{\omega_n} = \sqrt{\frac{(\omega_n)^2 - a^2}{\omega_n^2}} = \sqrt{\frac{(14.14)^2 - (0.5)^2}{14.14^2}}$$

$$\dots \left(\because \omega_d = \sqrt{(\omega_n)^2 - a^2} \right)$$

$$= 0.999 \quad \text{Ans.}$$

7. A machine of mass 75 kg is mounted on springs and is fitted with a dashpot to damp out vibrations. There are three springs each of stiffness 10 N/mm and it is found that the amplitude of vibration diminishes from 38.4 mm to 6.4 mm in two complete oscillations. Assuming that the damping force varies as the velocity, determine: 1. the resistance of the dashpot at unit velocity ; 2. the ratio of the frequency of the damped vibration to the frequency of the undamped vibration ; and 3. the periodic time of the damped vibration.

Solution. Given : $m = 75 \text{ kg}$; $s = 10 \text{ N/mm} = 10 \times 10^3 \text{ N/m}$; $x_1 = 38.4 \text{ mm} = 0.0384 \text{ m}$;
 $x_3 = 6.4 \text{ mm} = 0.0064 \text{ m}$

Since the stiffness of each spring is $10 \times 10^3 \text{ N/m}$ and there are 3 springs, therefore total stiffness,

$$s = 3 \times 10 \times 10^3 = 30 \times 10^3 \text{ N/m}$$

We know that natural circular frequency of motion,

$$\omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{30 \times 10^3}{75}} = 20 \text{ rad/s}$$

1. Resistance of the dashpot at unit velocity

Let c = Resistance of the dashpot in newtons at unit velocity i.e. in N/m/s,

x_2 = Amplitude after one complete oscillation in metres, and

x_3 = Amplitude after two complete oscillations in metres.

We know that $\frac{x_1}{x_2} = \frac{x_2}{x_3}$

$$\therefore \left(\frac{x_1}{x_2} \right)^2 = \frac{x_1}{x_3} \quad \dots \left[\because \frac{x_1}{x_3} = \frac{x_1}{x_2} \times \frac{x_2}{x_3} = \frac{x_1}{x_2} \times \frac{x_2}{x_2} = \left(\frac{x_1}{x_2} \right)^2 \right]$$

or
$$\frac{x_1}{x_2} = \left(\frac{x_1}{x_3} \right)^{1/2} = \left(\frac{0.0384}{0.0064} \right)^{1/2} = 2.45$$

We also know that

$$\log_e \left(\frac{x_1}{x_2} \right) = a \times \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

$$\log_e 2.45 = a \times \frac{2\pi}{\sqrt{(20)^2 - a^2}}$$

$$0.8951 = \frac{a \times 2\pi}{\sqrt{400 - a^2}} \quad \text{or} \quad 0.8 = \frac{a^2 \times 39.5}{400 - a^2} \quad \dots \text{(Squaring both sides)}$$

$$\therefore a^2 = 7.94 \quad \text{or} \quad a = 2.8$$

We know that $a = c / 2m$

$$\therefore c = a \times 2m = 2.8 \times 2 \times 75 = 420 \text{ N/m/s} \quad \text{Ans.}$$

2. Ratio of the frequency of the damped vibration to the frequency of undamped vibration

Let f_{n1} = Frequency of damped vibration = $\frac{\omega_d}{2\pi}$

f_{n2} = Frequency of undamped vibration = $\frac{\omega_n}{2\pi}$

$\therefore \frac{f_{n1}}{f_{n2}} = \frac{\omega_d}{2\pi} \times \frac{2\pi}{\omega_n} = \frac{\omega_d}{\omega_n} = \frac{\sqrt{(\omega_n)^2 - a^2}}{\omega_n} = \frac{\sqrt{(20)^2 - (2.8)^2}}{20} = 0.99$ **Ans.**

3. Periodic time of damped vibration

We know that periodic time of damped vibration

$$= \frac{2\pi}{\omega_d} = \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}} = \frac{2\pi}{\sqrt{(20)^2 - (2.8)^2}} = 0.32 \text{ s}$$
 Ans.

8. The mass of a single degree damped vibrating system is 7.5 kg and makes 24 free oscillations in 14 seconds when disturbed from its equilibrium position. The amplitude of vibration reduces to 0.25 of its initial value after five oscillations. Determine : 1. stiffness of the spring, 2. logarithmic decrement, and 3. damping factor, i.e. the ratio of the system damping to critical damping.

Solution. Given : $m = 7.5$ kg

Since 24 oscillations are made in 14 seconds, therefore frequency of free vibrations,

$$f_n = 24/14 = 1.7$$

and

$$\omega_n = 2\pi \times f_n = 2\pi \times 1.7 = 10.7 \text{ rad/s}$$

1. Stiffness of the spring

Let s = Stiffness of the spring in N/m.

We know that $(\omega_n)^2 = s/m$ or $s = (\omega_n)^2 m = (10.7)^2 7.5 = 860$ N/m **Ans.**

2. Logarithmic decrement

Let x_1 = Initial amplitude,

x_6 = Final amplitude after five oscillations = $0.25 x_1$... (Given)

$\therefore \frac{x_1}{x_6} = \frac{x_1}{x_2} \times \frac{x_2}{x_3} \times \frac{x_3}{x_4} \times \frac{x_4}{x_5} \times \frac{x_5}{x_6} = \left(\frac{x_1}{x_2} \right)^5 \dots \left[\because \frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \frac{x_4}{x_5} = \frac{x_5}{x_6} \right]$

or $\frac{x_1}{x_2} = \left(\frac{x_1}{x_6} \right)^{1/5} = \left(\frac{x_1}{0.25 x_1} \right)^{1/5} = (4)^{1/5} = 1.32$

We know that logarithmic decrement,

$$\delta = \log_e \left(\frac{x_1}{x_2} \right) = \log_e 1.32 = 0.28$$
 Ans.

3. Damping factor

Let

c = Damping coefficient for the actual system, and

c_c = Damping coefficient for the critical damped system.

We know that logarithmic decrement (δ),

$$0.28 = \frac{a \times 2\pi}{\sqrt{(\omega_n)^2 - a^2}} = \frac{a \times 2\pi}{\sqrt{(10.7)^2 - a^2}}$$

$$0.0784 = \frac{a^2 \times 39.5}{114.5 - a^2} \quad \dots \text{(Squaring both sides)}$$

$$8.977 - 0.0784 a^2 = 39.5 a^2 \quad \text{or} \quad a^2 = 0.227 \quad \text{or} \quad a = 0.476$$

We know that $a = c / 2m$ or $c = a \times 2m = 0.476 \times 2 \times 7.5 = 7.2 \text{ N/m/s}$ **Ans.**

and

$$c_c = 2m\omega_n = 2 \times 7.5 \times 10.7 = 160.5 \text{ N/m/s} \text{ **Ans.**}$$

$$\therefore \text{Damping factor} = c/c_c = 7.2 / 160.5 = 0.045 \text{ **Ans.**}$$

FREQUENCY OF UNDER DAMPED FORCED VIBRATIONS

$$F_x = F \cos \omega t$$

where

F = Static force, and
 ω = Angular velocity of
the periodic disturbing
force.

The equation of motion is given as,

$$m \times \frac{d^2 x}{dt^2} = -c \times \frac{dx}{dt} - s.x + F \cos \omega t$$

or

$$m \times \frac{d^2 x}{dt^2} + c \times \frac{dx}{dt} + s.x = F \cos \omega t$$

Maximum displacement or the amplitude of forced vibration,

$$x_{max} = \frac{F}{\sqrt{c^2 \omega^2 + (s - m \omega^2)^2}}$$

$$x_{max} = \frac{F/s}{\sqrt{\frac{c^2 \omega^2}{s^2} + \frac{(s - m \omega^2)^2}{s^2}}}$$

$$= \frac{x_o}{\sqrt{\frac{c^2 \omega^2}{s^2} + \left(1 - \frac{m \omega^2}{s}\right)^2}}$$

... (Substituting $F/s = x_o$)

$$(\omega_n)^2 = s/m$$

$$x_{max} = \frac{x_o}{\sqrt{\frac{c^2 \omega^2}{s^2} + \left(1 - \frac{\omega^2}{(\omega_n)^2}\right)^2}}$$

$\therefore$

When damping is negligible, then $c = 0$.

$$x_{max} = \frac{x_o}{1 - \frac{\omega^2}{(\omega_n)^2}} = \frac{x_o (\omega_n)^2}{(\omega_n)^2 - \omega^2} = \frac{x_o \times s/m}{(\omega_n)^2 - \omega^2} \quad \dots \left[\because (\omega_n)^2 = s/m \right]$$

$$= \frac{F}{m \left[(\omega_n)^2 - \omega^2 \right]} \quad \dots (\because F = x_o \cdot s)$$

At resonance $\omega = \omega_n$. Therefore the angular speed at which the resonance occurs is

$$\omega = \omega_n = \sqrt{\frac{s}{m}} \text{ rad/s}$$

$$x_{max} = x_o \times \frac{s}{c \cdot \omega_n}$$

Magnification factor or dynamic magnifier,

$$D = \frac{x_{max}}{x_o} = \frac{1}{\sqrt{\frac{c^2 \omega^2}{s^2} + \left(1 - \frac{\omega^2}{(\omega_n)^2}\right)^2}}$$

Notes: 1. If there is no damping (i.e. if the vibration is undamped), then $c = 0$. In that case, magnification factor,

$$D = \frac{x_{max}}{x_o} = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{(\omega_n)^2}\right)^2}} = \frac{(\omega_n)^2}{(\omega_n)^2 - \omega^2}$$

2. At resonance, $\omega = \omega_n$. Therefore magnification factor,

$$D = \frac{x_{max}}{x_o} = \frac{s}{c \cdot \omega_n}$$

PROBLEMS ON UNDER DAMPED FORCED VIBRATIONS:

1. A single cylinder vertical petrol engine of total mass 300 kg is mounted upon a steel chassis frame and causes a vertical static deflection of 2 mm. The reciprocating parts of the engine has a mass of 20 kg and move through a vertical stroke of 150 mm with simple harmonic motion. A dashpot is provided whose damping resistance is directly proportional to the velocity and amounts to 1.5 kN per metre per second. Considering that the steady state of vibration is reached ; determine : 1. the amplitude of forced vibrations, when the driving shaft of the engine rotates at 480 r.p.m., and 2. the speed of the driving shaft at which resonance will occur.

Solution : Given. $m = 300$ kg; $\delta = 2$ mm $= 2 \times 10^{-3}$ m ; $m_1 = 20$ kg ; $l = 150$ mm $= 0.15$ m ; $c = 1.5$ kN/m/s $= 1500$ N/m/s ; $N = 480$ r.p.m. or $\omega = 2\pi \times 480 / 60 = 50.3$ rad/s

1. Amplitude of the forced vibrations

We know that stiffness of the frame,

$$s = m.g / \delta = 300 \times 9.81 / 2 \times 10^{-3} = 1.47 \times 10^6 \text{ N/m}$$

Since the length of stroke (l) = 150 mm = 0.15 m, therefore radius of crank,

$$r = l / 2 = 0.15 / 2 = 0.075 \text{ m}$$

We know that the centrifugal force due to the reciprocating parts or the static force,

$$F = m_1 \omega^2 r = 20 (50.3)^2 0.075 = 3795 \text{ N}$$

$\therefore$ Amplitude of the forced vibration (maximum),

$$\begin{aligned} x_{max} &= \frac{F}{\sqrt{c^2 \omega^2 + (s - m \omega^2)^2}} \\ &= \frac{3795}{\sqrt{(1500)^2 (50.3)^2 + [1.47 \times 10^6 - 300 (50.3)^2]^2}} \\ &= \frac{3795}{\sqrt{5.7 \times 10^9 + 500 \times 10^9}} = \frac{3795}{710 \times 10^3} = 5.3 \times 10^{-3} \text{ m} \\ &= 5.3 \text{ mm} \text{ Ans.} \end{aligned}$$

2. Speed of the driving shaft at which the resonance occurs

Let

N = Speed of the driving shaft at which the resonance occurs in r.p.m.

We know that the angular speed at which the resonance occurs,

$$\omega = \omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{1.47 \times 10^6}{300}} = 70 \text{ rad/s}$$

$\therefore$

$$N = \omega \times 60 / 2\pi = 70 \times 60 / 2\pi = 668.4 \text{ r.p.m. Ans.}$$

2. A mass of 10 kg is suspended from one end of a helical spring, the other end being fixed. The stiffness of the spring is 10 N/mm. The viscous damping causes the amplitude to decrease to one-tenth of the initial value in four complete oscillations. If a periodic force of $150 \cos 50 t$ N is applied at the mass in the vertical direction, find the amplitude of the forced vibrations. What is its value of resonance?

Solution. Given : $m = 10 \text{ kg}$; $s = 10 \text{ N/mm} = 10 \times 10^3 \text{ N/m}$; $x_5 = \frac{x_1}{10}$

Since the periodic force, $F_x = F \cos \omega t = 150 \cos 50 t$, therefore

Static force, $F = 150 \text{ N}$

and angular velocity of the periodic disturbing force,

$$\omega = 50 \text{ rad/s}$$

We know that angular speed or natural circular frequency of free vibrations,

$$\omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{10 \times 10^3}{10}} = 31.6 \text{ rad/s}$$

Amplitude of the forced vibrations

Since the amplitude decreases to 1/10th of the initial value in four complete oscillations, therefore, the ratio of initial amplitude (x_1) to the final amplitude after four complete oscillations (x_5) is given by

$$\frac{x_1}{x_5} = \frac{x_1}{x_2} \times \frac{x_2}{x_3} \times \frac{x_3}{x_4} \times \frac{x_4}{x_5} = \left(\frac{x_1}{x_2} \right)^4 \quad \dots \left(\because \frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \frac{x_4}{x_5} \right)$$

$$\therefore \frac{x_1}{x_2} = \left(\frac{x_1}{x_5} \right)^{1/4} = \left(\frac{x_1}{x_1/10} \right)^{1/4} = (10)^{1/4} = 1.78 \quad \dots \left(x_5 = \frac{x_1}{10} \right)$$

We know that

$$\log_e \left(\frac{x_1}{x_2} \right) = a \times \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

$$\log_e 1.78 = a \times \frac{2\pi}{\sqrt{(31.6)^2 - a^2}} \quad \text{or} \quad 0.576 = \frac{a \times 2\pi}{\sqrt{1000 - a^2}}$$

Squaring both sides and rearranging,

$$39.832 a^2 = 332 \quad \text{or} \quad a^2 = 8.335 \quad \text{or} \quad a = 2.887$$

We know that $a = c/2m$ or $c = a \times 2m = 2.887 \times 2 \times 10 = 57.74 \text{ N/m/s}$
and deflection of the system produced by the static force F ,

$$x_o = F/s = 150/10 \times 10^3 = 0.015 \text{ m}$$

We know that amplitude of the forced vibrations,

$$x_{max} = \frac{x_o}{\sqrt{\frac{c^2 \omega^2}{s^2} + \left[1 - \frac{\omega^2}{(\omega_n)^2} \right]^2}}$$

$$= \frac{0.015}{\sqrt{\frac{(57.74)^2(50)^2}{(10 \times 10^3)^2} + \left[1 - \left(\frac{50}{31.6}\right)^2\right]^2}} = \frac{0.015}{\sqrt{0.083 + 2.25}}$$

$$= \frac{0.015}{1.53} = 9.8 \times 10^{-3} \text{ m} = 9.8 \text{ mm} \text{ Ans.}$$

Amplitude of forced vibrations at resonance

We know that amplitude of forced vibrations at resonance,

$$x_{\max} = x_0 \times \frac{s}{c \cdot \omega_n} = 0.015 \times \frac{10 \times 10^3}{57.54 \times 31.6} = 0.0822 \text{ m} = 82.2 \text{ mm} \text{ Ans.}$$

3. A body of mass 20 kg is suspended from a spring which deflects 15 mm under this load. Calculate the frequency of free vibrations and verify that a viscous damping force amounting to approximately 1000 N at a speed of 1 m/s is just-sufficient to make the motion aperiodic. If when damped to this extent, the body is subjected to a disturbing force with a maximum value of 125 N making 8 cycles/s, find the amplitude of the ultimate motion.

Solution . Given : $m = 20 \text{ kg}$; $\delta = 15 \text{ mm} = 0.015 \text{ m}$; $c = 1000 \text{ N/m/s}$; $F = 125 \text{ N}$; $f = 8 \text{ cycles/s}$

Frequency of free vibrations

We know that frequency of free vibrations,

$$f_n = \frac{1}{2\pi} \sqrt{\frac{g}{\delta}} = \frac{1}{2\pi} \sqrt{\frac{9.81}{0.015}} = 4.07 \text{ Hz} \text{ Ans.}$$

The critical damping to make the motion aperiodic is such that damped frequency is zero, i.e.

$$\left(\frac{c}{2m}\right)^2 = \frac{s}{m}$$

$$\therefore c = \sqrt{\frac{s}{m} \times 4m^2} = \sqrt{4sm} = \sqrt{4 \times \frac{m \cdot g}{\delta} \times m} \quad \dots \left(\because s = \frac{m \cdot g}{\delta}\right)$$

$$= \sqrt{4 \times \frac{20 \times 9.81}{0.015} \times 20} = 1023 \text{ N/m/s}$$

This means that the viscous damping force is 1023 N at a speed of 1 m/s. Therefore a viscous damping force amounting to approximately 1000 N at a speed of 1 m/s is just sufficient to make the motion aperiodic. **Ans.**

Amplitude of ultimate motion

We know that angular speed of forced vibration,

$$\omega = 2\pi \times f = 2\pi \times 8 = 50.3 \text{ rad/s}$$

and stiffness of the spring, $s = m \cdot g / \delta = 20 \times 9.81 / 0.015 = 13.1 \times 10^3 \text{ N/m}$

$\therefore$ Amplitude of ultimate motion *i.e.* maximum amplitude of forced vibration,

$$\begin{aligned} x_{max} &= \frac{F}{\sqrt{c^2 \cdot \omega^2 + (s - m \cdot \omega^2)^2}} \\ &= \frac{125}{\sqrt{(1023)^2 (50.3)^2 + [13.1 \times 10^3 - 20(50.3)^2]^2}} \\ &= \frac{125}{\sqrt{2600 \times 10^6 + 1406 \times 10^6}} = \frac{125}{63.7 \times 10^3} = 1.96 \times 10^{-3} \text{ m} \\ &= 1.96 \text{ mm} \text{ Ans.} \end{aligned}$$

4. A machine part of mass 2 kg vibrates in a viscous medium. Determine the damping coefficient when a harmonic exciting force of 25 N results in a resonant amplitude of 12.5 mm with a period of 0.2 second. If the system is excited by a harmonic force of frequency 4 Hz what will be the percentage increase in the amplitude of vibration when damper is removed as compared with that with damping.

Solution . Given : $m = 2 \text{ kg}$; $F = 25 \text{ N}$; Resonant $x_{max} = 12.5 \text{ mm} = 0.0125 \text{ m}$;
 $t_p = 0.2 \text{ s}$; $f = 4 \text{ Hz}$

Damping coefficient

Let c = Damping coefficient in N/m/s.

We know that natural circular frequency of the existing force,

$$\omega_n = 2\pi / t_p = 2\pi / 0.2 = 31.42 \text{ rad/s}$$

We also know that the maximum amplitude of vibration at resonance (x_{max}),

$$0.0125 = \frac{F}{c \cdot \omega_n} = \frac{25}{c \times 31.42} = \frac{0.796}{c} \text{ or } c = 63.7 \text{ N/m/s} \text{ Ans.}$$

Percentage increase in amplitude

Since the system is excited by a harmonic force of frequency (f) = 4 Hz, therefore corresponding circular frequency

$$\omega = 2\pi \times f = 2\pi \times 4 = 25.14 \text{ rad/s}$$

We know that maximum amplitude of vibration with damping,

$$x_{max} = \frac{F}{\sqrt{c^2 \cdot \omega^2 + (s - m \cdot \omega^2)^2}}$$

$$= \frac{25}{\sqrt{(63.7)^2 (25.14)^2 + [2(31.42)^2 - 2(25.14)^2]^2}}$$

$$\dots \left[\because (\omega_n)^2 = s/m \text{ or } s = m(\omega_n)^2 \right]$$

$$= \frac{25}{\sqrt{2.56 \times 10^6 + 0.5 \times 10^6}} = \frac{25}{1749} = 0.0143 \text{ m} = 14.3 \text{ mm}$$

and the maximum amplitude of vibration when damper is removed,

$$x_{max} = \frac{F}{m[(\omega_n)^2 - \omega^2]} = \frac{25}{2[(31.42)^2 - (25.14)^2]} = \frac{25}{710} = 0.0352 \text{ m}$$

$$= 35.2 \text{ mm}$$

$\therefore$ Percentage increase in amplitude

$$= \frac{35.2 - 14.3}{14.3} = 1.46 \text{ or } 146\% \text{ Ans.}$$

5. The time of free vibration of a mass hung from the end of a helical spring is 0.8 second. When the mass is stationary, the upper end is made to move upwards with a displacement y metre such that $y = 0.018 \sin 2\pi t$, where t is the time in seconds measured from the beginning of the motion. Neglecting the mass of the spring and any damping effects, determine the vertical distance through which the mass is moved in the first 0.3 second.

Solution. Given : $t_p = 0.8 \text{ s}$; $y = 0.018 \sin 2\pi t$

Let m = Mass hung to the spring in kg, and
 s = Stiffness of the spring in N/m.

We know that time period of free vibrations (t_p),

$$0.8 = 2\pi \sqrt{\frac{m}{s}} \quad \text{or} \quad \frac{m}{s} = \left(\frac{0.8}{2\pi} \right)^2 = 0.0162$$

If x metres is the upward displacement of mass m from its equilibrium position after time t seconds, the equation of motion is given by

$$m \times \frac{d^2 x}{dt^2} + s(y - x) = 0 \quad \text{or} \quad \frac{m}{s} \times \frac{d^2 x}{dt^2} + x - y = 0.018 \sin 2\pi t$$

The solution of this differential equation is

$$x = A \sin \sqrt{\frac{s}{m}} \times t + B \cos \sqrt{\frac{s}{m}} \times t + \frac{0.018 \sin 2\pi t}{1 - \left(\frac{2\pi}{\sqrt{s/m}} \right)^2}$$

$\dots$ (where A and B are constants)

$$= A \sin \frac{t}{\sqrt{0.0162}} + B \cos \frac{t}{\sqrt{0.0162}} + \frac{0.018 \sin 2\pi t}{1 - 4\pi^2 \times 0.0162}$$

$$= A \sin 7.85t + B \cos 7.85t + 0.05 \sin 2\pi t \quad \dots (i)$$

Now when $t = 0$, $x = 0$, then from equation (i), $B = 0$

Again when $t = 0$, $dx/dt = 0$.

Therefore differentiating equation (i) and equating to zero, we have

$$dx/dt = 7.85A \cos 7.85t + 0.05 \times 2\pi \cos 2\pi t = 0 \quad \dots (\because B = 0)$$

or $7.85A \cos 7.85t = -0.05 \times 2\pi \cos 2\pi t$

$$\therefore A = -0.05 \times 2\pi / 7.85 = -0.04 \quad \dots (\because t = 0)$$

Now the equation (i) becomes

$$x = -0.04 \sin 7.85t + 0.05 \sin 2\pi t \quad \dots (\because B = 0) \quad \dots (ii)$$

$\therefore$ Vertical distance through which the mass is moved in the first 0.3 second (i.e. when $t = 0.3$ s),

$$= -0.04 \sin (7.85 \times 0.3) + 0.05 \sin (2\pi \times 0.3)$$

$\dots$ [Substituting $t = 0.3$ in equation (ii)]

$$= -0.04 \times 0.708 + 0.05 \times 0.951 = -0.0283 + 0.0476 = 0.0193 \text{ m}$$

$$= 19.3 \text{ mm Ans.}$$

VIBRATION ISOLATION AND TRANSMISSIBILITY

$$F_T = \sqrt{(s.x_{max})^2 + (c.\omega.x_{max})^2}$$

$$= x_{max} \sqrt{s^2 + c^2.\omega^2}$$

$\therefore$ Transmissibility ratio,

$$\epsilon = \frac{F_T}{F} = \frac{x_{max} \sqrt{s^2 + c^2.\omega^2}}{F}$$

$$\epsilon = \frac{\sqrt{1 + \left(\frac{2c.\omega}{c_c.\omega_n} \right)^2}}{\sqrt{\left(\frac{2c.\omega}{c_c.\omega_n} \right)^2 + \left(1 - \frac{\omega^2}{(\omega_n)^2} \right)^2}}$$

When the damper is not provided, then $c = 0$, and

$$\epsilon = \frac{1}{1 - (\omega/\omega_n)^2}$$

PROBLEMS ON VIBRATION AND TRANSMISSIBILITY:

1. The mass of an electric motor is 120 kg and it runs at 1500 r.p.m. The armature mass is 35 kg and its C.G. lies 0.5 mm from the axis of rotation. The motor is mounted on five springs of negligible damping so that the force transmitted is one-eleventh of the impressed force. Assume that the mass of the motor is equally distributed among the five springs. Determine: 1. stiffness of each spring; 2. dynamic force transmitted to the base at the operating speed; and 3. natural frequency of the system.

Solution. Given $m_1 = 120$ kg; $m_2 = 35$ kg; $r = 0.5$ mm $= 5 \times 10^{-4}$ m; $\epsilon = 1/11$; $N = 1500$ r.p.m. or $\omega = 2\pi \times 1500 / 60 = 157.1$ rad/s;

1. Stiffness of each spring

Let s = Combined stiffness of the spring in N-m, and
 ω_n = Natural circular frequency of vibration of the machine in rad/s.

We know that transmissibility ratio (ϵ),

$$\frac{1}{11} = \frac{1}{\left(\frac{\omega}{\omega_n} \right)^2 - 1} = \frac{(\omega_n)^2}{\omega^2 - (\omega_n)^2} = \frac{(\omega_n)^2}{(157.1)^2 - (\omega_n)^2}$$

or $(157.1)^2 - (\omega_n)^2 = 11(\omega_n)^2$ or $(\omega_n)^2 = 2057$ or $\omega_n = 45.35$ rad/s

We know that $\omega_n = \sqrt{s/m_1}$

$$s = m_1 (\omega_n)^2 = 120 \times 2057 = 246\,840 \text{ N/m}$$

Since these are five springs, therefore stiffness of each spring

$$= 246\,840 / 5 = 49\,368 \text{ N/m} \quad \text{Ans.}$$

2. Dynamic force transmitted to the base at the operating speed (i.e. 1500 r.p.m. or 157.1 rad/s)

We know that maximum unbalanced force on the motor due to armature mass,

$$F = m_2 \omega^2 \cdot r = 35 (157.1)^2 \cdot 5 \times 10^{-4} = 432 \text{ N}$$

$\therefore$ Dynamic force transmitted to the base,

$$F_T = \epsilon \cdot F = \frac{1}{11} \times 432 = 39.27 \text{ N} \quad \text{Ans.}$$

3. Natural frequency of the system

We have calculated above that the natural frequency of the system,

$$\omega_n = 45.35 \text{ rad/s} \quad \text{Ans.}$$

2. A machine has a mass of 100 kg and unbalanced reciprocating parts of mass 2 kg which move through a vertical stroke of 80 mm with simple harmonic motion. The machine is mounted on four springs, symmetrically arranged with respect to centre of mass, in such a way that the machine has one degree of freedom and can undergo vertical displacements only. Neglecting damping, calculate the combined stiffness of the spring in order that the force transmitted to the foundation is 1 / 25 th of the applied force, when the speed of rotation of machine crank shaft is 1000 r.p.m. When the machine is actually supported on the springs, it is found that the damping reduces the amplitude of successive free vibrations by 25%. Find : 1. the force transmitted to foundation at 1000 r.p.m., 2. the force transmitted to the foundation at resonance, and 3. the amplitude of the forced vibration of the machine at resonance.

Solution. Given : $m_1 = 100 \text{ kg}$; $m_2 = 2 \text{ kg}$; $l = 80 \text{ mm} = 0.08 \text{ m}$; $\varepsilon = 1/2$
 $N = 1000 \text{ r.p.m.}$ or $\omega = 2\pi \times 1000 / 60 = 104.7 \text{ rad/s}$

Combined stiffness of springs

Let s = Combined stiffness of springs in N/m, and
 ω_n = Natural circular frequency of vibration of the machine in rad/s
 We know that transmissibility ratio (ε),

$$\frac{1}{25} = \frac{1}{\left(\frac{\omega}{\omega_n}\right)^2 - 1} = \frac{(\omega_n)^2}{\omega^2 - (\omega_n)^2} = \frac{(\omega_n)^2}{(104.7)^2 - (\omega_n)^2}$$

or $(104.7)^2 - (\omega_n)^2 = 25(\omega_n)^2$ or $(\omega_n)^2 = 421.6$ or $\omega_n = 20.5 \text{ rad/s}$

We know that $\omega_n = \sqrt{s/m_1}$

$\therefore s = m_1 (\omega_n)^2 = 100 \times 421.6 = 42\,160 \text{ N/m}$ **Ans.**

1. Force transmitted to the foundation at 1000 r.p.m.

Let F_T = Force transmitted, and
 x_1 = Initial amplitude of vibration.

Since the damping reduces the amplitude of successive free vibrations by 25%, therefore final amplitude of vibration,

$$x_2 = 0.75 x_1$$

We know that

$$\log_e \left(\frac{x_1}{x_2} \right) = \frac{a \times 2\pi}{\sqrt{(\omega_n)^2 - a^2}} \quad \text{or} \quad \log_e \left(\frac{x_1}{0.75x_1} \right) = \frac{a \times 2\pi}{\sqrt{421.6 - a^2}}$$

Squaring both sides,

$$(0.2877)^2 = \frac{a^2 \times 4\pi^2}{421.6 - a^2} \quad \text{or} \quad 0.083 = \frac{39.5 a^2}{421.6 - a^2}$$

$$\dots \left[\because \log_e \left(\frac{1}{0.75} \right) = \log_e 1.333 = 0.2877 \right]$$

$$35 - 0.083 a^2 = 39.5 a^2 \quad \text{or} \quad a^2 = 0.884 \quad \text{or} \quad a = 0.94$$

We know that damping coefficient or damping force per unit velocity,

$$c = a \times 2m_1 = 0.94 \times 2 \times 100 = 188 \text{ N/m/s}$$

and critical damping coefficient,

$$c_c = 2m_1\omega_n = 2 \times 100 \times 20.5 = 4100 \text{ N/m/s}$$

$\therefore$ Actual value of transmissibility ratio,

$$\begin{aligned} \varepsilon &= \frac{\sqrt{1 + \left(\frac{2c\omega}{c_c\omega_n}\right)^2}}{\sqrt{\left(\frac{2c\omega}{c_c\omega_n}\right)^2 + \left[1 - \frac{\omega^2}{(\omega_n)^2}\right]^2}} \\ &= \frac{\sqrt{1 + \left(\frac{2 \times 188 \times 104.7}{4100 \times 20.5}\right)^2}}{\sqrt{\left(\frac{2 \times 188 \times 104.7}{4100 \times 20.5}\right)^2 + \left[1 - \left(\frac{104.7}{20.5}\right)^2\right]^2}} = \frac{\sqrt{1 + 0.22}}{\sqrt{0.22 + 629}} \\ &= \frac{1.104}{25.08} = 0.044 \end{aligned}$$

We know that the maximum unbalanced force on the machine due to reciprocating parts,

$$F = m_2 \omega^2 r = 2 (104.7)^2 (0.08/2) = 877 \text{ N} \quad \dots (\because r = l/2)$$

$\therefore$ Force transmitted to the foundation,

$$F_T = \varepsilon F = 0.044 \times 877 = 38.6 \text{ N Ans.} \quad \dots (\because \varepsilon = F_T/F)$$

2. Force transmitted to the foundation at resonance

Since at resonance, $\omega = \omega_n$, therefore transmissibility ratio,

$$\varepsilon = \frac{\sqrt{1 + \left(\frac{2c}{c_c}\right)^2}}{\sqrt{\left(\frac{2c}{c_c}\right)^2}} = \frac{\sqrt{1 + \left(\frac{2 \times 188}{4100}\right)^2}}{\sqrt{\left(\frac{2 \times 188}{4100}\right)^2}} = \frac{\sqrt{1 + 0.0084}}{0.092} = 10.92$$

and maximum unbalanced force on the machine due to reciprocating parts at resonance speed ω_n ,

$$F = m_2 (\omega_n)^2 r = 2 (20.5)^2 (0.08/2) = 33.6 \text{ N} \quad \dots (\because r = l/2)$$

$\therefore$ Force transmitted to the foundation at resonance,

$$F_T = \varepsilon F = 10.92 \times 33.6 = 367 \text{ N Ans.}$$

3. Amplitude of the forced vibration of the machine at resonance

We know that amplitude of the forced vibration at resonance

$$\begin{aligned} &= \frac{\text{Force transmitted at resonance}}{\text{Combined stiffness}} = \frac{367}{42160} = 8.7 \times 10^{-3} \text{ m} \\ &= 8.7 \text{ mm Ans.} \end{aligned}$$

3. A single-cylinder engine of total mass 200 kg is to be mounted on an elastic support which permits vibratory movement in vertical direction only. The mass of the piston is 3.5 kg and has a vertical reciprocating motion which may be assumed simple harmonic with a stroke of 150 mm. It is desired that the maximum vibratory force transmitted through the elastic support to the foundation shall be 600 N when the engine speed is 800 r.p.m. and less than this at all higher speeds. 1. Find the necessary stiffness of the elastic support, and the amplitude of vibration at 800 r.p.m., and 2. If the engine speed is reduced below 800 r.p.m. at what speed will the transmitted force again becomes 600 N?

Solution. Given : $m_1 = 200 \text{ kg}$; $m_2 = 3.5 \text{ kg}$; $l = 150 \text{ mm} = 0.15 \text{ m}$ or $r = l/2 = 0.075 \text{ m}$;

$F_T = 600 \text{ N}$; $N = 800 \text{ r.p.m.}$ or $\omega = 2\pi \times 800 / 60 = 83.8 \text{ rad/s}$

We know that the disturbing force at 800 r.p.m.,

$F = \text{Centrifugal force on the piston}$

$$= m_2 \omega^2 r = 3.5 (83.8)^2 0.075 = 1843 \text{ N}$$

1. Stiffness of elastic support and amplitude of vibration

Let $s = \text{Stiffness of elastic support in N/m, and}$

$x_{max} = \text{Max. amplitude of vibration in metres.}$

Since the max. vibratory force transmitted to the foundation is equal to the force on the elastic support (neglecting damping), therefore

Max. vibratory force transmitted to the foundation,

$F_T = \text{Force on the elastic support}$

$= \text{Stiffness of elastic support} \times \text{Max. amplitude of vibration}$

$$= s \times x_{max} = s \times \frac{F}{m[\omega^2 - (\omega_n)^2]}$$

$$= s \times \frac{F}{m\left(\omega^2 - \frac{s}{m}\right)} = \frac{F s}{m \omega^2 - s} \quad \dots \left[\because (\omega_n)^2 = \frac{s}{m} \right]$$

$$\therefore 600 = \frac{1843 \times s}{200(83.8)^2 - s} = \frac{1843 s}{1.4 \times 10^6 - s} \quad \dots (\text{Substituting } m = m_1)$$

or $840 \times 10^6 - 600 s = 1843 s$

$$\therefore s = 0.344 \times 10^6 = 344 \times 10^3 \text{ N/m Ans.}$$

and maximum amplitude of vibration,

$$\begin{aligned} x_{max} &= \frac{F}{m \omega^2 - s} = \frac{1843}{200(83.8)^2 - 344 \times 10^3} = \frac{1843}{1056 \times 10^3} \text{ m} \\ &= 1.745 \times 10^{-3} \text{ m} = 1.745 \text{ mm Ans.} \end{aligned}$$

2. Speed at the which the transmitted force again becomes 600 N

The transmitted force will rise as the speed of the engine falls and passes through resonance. There will be a speed below resonance at which the transmitted force will again equal to 600 N. Let this speed be ω_1 rad/s (or N_1 r.p.m.).

$$\therefore \text{Disturbing force, } F = m_2 (\omega_1)^2 r = 3.5 (\omega_1)^2 0.075 = 0.2625 (\omega_1)^2 \text{ N}$$

Since the engine speed is reduced below $N_1 = 800$ r.p.m., therefore in this case, max. amplitude of vibration,

$$x_{max} = \frac{F}{m \left[(\omega_n)^2 - (\omega_1)^2 \right]} = \frac{F}{m \left[\frac{s}{m} - (\omega_1)^2 \right]} = \frac{F}{s - m (\omega_1)^2}$$

and $\text{Force transmitted} = s \times \frac{F}{s - m (\omega_1)^2}$

$$\therefore 600 = 344 \times 10^3 \times \frac{0.2625 (\omega_1)^2}{344 \times 10^3 - 200 (\omega_1)^2} = \frac{90.3 \times 10^3 (\omega_1)^2}{344 \times 10^3 - 200 (\omega_1)^2}$$

... (Substituting $m = m_1$)

$$206.4 \times 10^6 - 120 \times 10^3 (\omega_1)^2 = 90.3 \times 10^3 (\omega_1)^2 \quad \text{or} \quad (\omega_1)^2 = 981$$

$$\therefore \omega_1 = 31.32 \text{ rad/s or } N_1 = 31.32 \times 60 / 2\pi = 299 \text{ r.p.m. } \textbf{Ans.}$$
